Antique Radio as Bluetooth Speaker

by **dermbrian** on June 18, 2014

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Intro: Antique Radio as Bluetooth Speaker

I've always felt an affection for old radios. The fact that the United States of America used to lead the world in the manufacturing of these consumer items is kind of hard to believe in the Year of Our Lord 2014, but it's true.

In the mid 1990's, I purchased a kind of unique radio from a gentleman in Florida. It's a 1929 lowboy cabinet that surely originally housed a Sparton 931 radio chassis (google images is your friend). The cabinet itself is in pretty good shape, but some of the ornate woodwork mouldings are missing. Someone did a reasonable job of fashioning some items that give a sense of the original items without being too distracting. The radio chassis itself is a Philco 70 cathedral radio chassis, manufactured in 1931. The radio worked when it was shipped to me, but UPS laid the crate on its side and left it on our patio. That caused the chassis shelf to break free, smashing two tubes and tearing the orignal electromagnetic speaker to shreds. The seller sent me a couple of replacement tubes, and I researched replacing the electromagnetic speaker with a permanent magnet modern speaker. I obtained a schematic and was eventually able to get the radio working.

Step 1: Initial kludge

"Working" is a relative term. The radio pre-dates Automatic Gain Control, so tuning through the AM band was an exercise in how fast you could lower the volume when you came to a local powerhouse station, and crank it up again as you try to find a fainter signal. The fidelity was not very good, probably a combination of my repairs and the radio technology itself. After fixing it, the radio served as a nightstand in our bedroom. Over the years, I played around with using the radio with a set of computer speakers in the speaker hole and playing MP3's or attaching another radio to that setup to listen to a broadcast. I put a nightlight behind the dial to light it up. It still didn't get much use.

Step 2: Eureka moment

Recently, I ran across a type of speaker I didn't even know existed: Ceiling speakers with dual voice coils, allowing the projection of both left and right stereo channels in one speaker. A typical size is 6.5 inches, which is just about ideal for the speaker cutout in my radio cabinet. This would allow me to use a low-cost class T-Amp (I have three of the original Sonic Impact T-Amps...) to amplify the signal source and drive the dual-channel speaker.

Bluetooth audio has come a long way and was not available during my first attempted mods to this radio cabinet. I did some reading on Bluetooth and fidelity and quickly learned that I wanted to incorporate a Bluetooth receiver into the cabinet, and that one capable of supporting the Apt-x codec would be best. My Galaxy Note 2 phone contains my entire music collection on a 128gb microSD card and supports Apt-x Bluetooth. I was all set to convert the radio cabinet/radio to a big Bluetooth speaker system.

The Bluetooth receiver I chose was this one, about $35.00 purchased from Amazon.com:

Brightech - BrightPlay Home HD™ Bluetooth 4.0 Music Receiver

Step 3: Modifying the radio, respectfully

Now that I knew I had a concept that would really let this radio "live" again, I decided to put a little more effort in it. My first motto was **"Do No Harm"** to this 83 year old radio chassis, beyond the harm already done by UPS in the original delivery. I knew I wanted to be able to do the following:

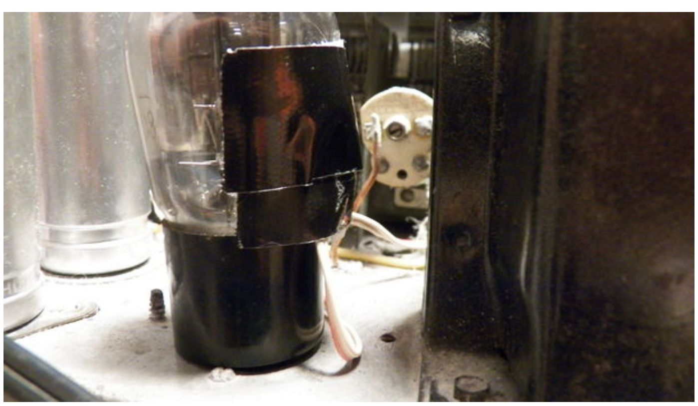
1. Power the bluetooth receiver and T-amp by using the power switch on the radio.

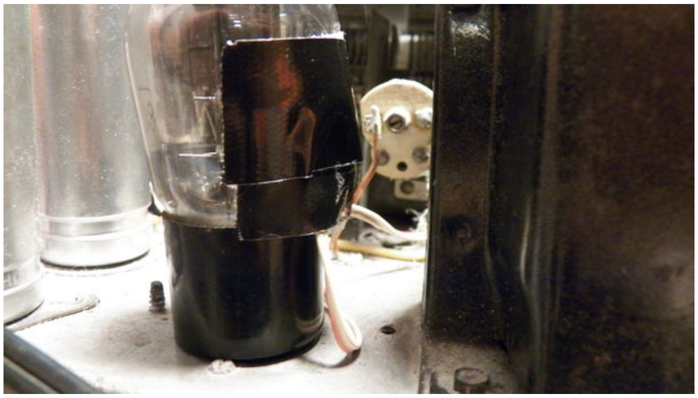
2. Light up the dial using a yellow LED.

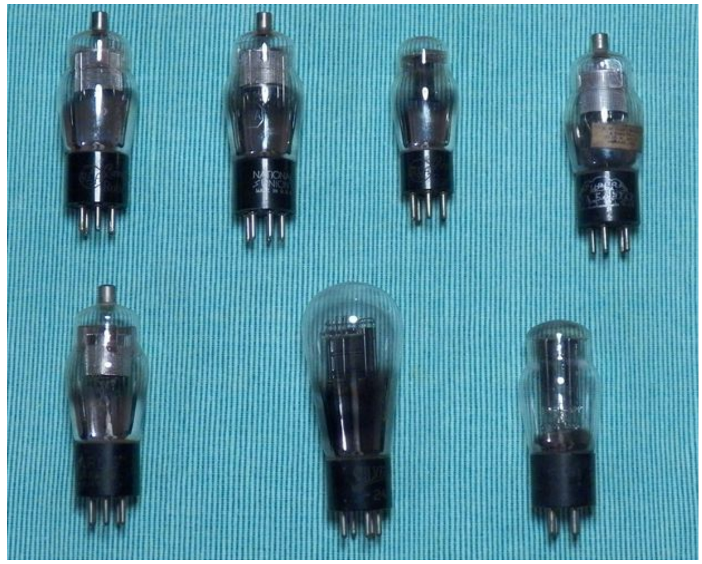
3. Make the tubes glow as if they are in use.

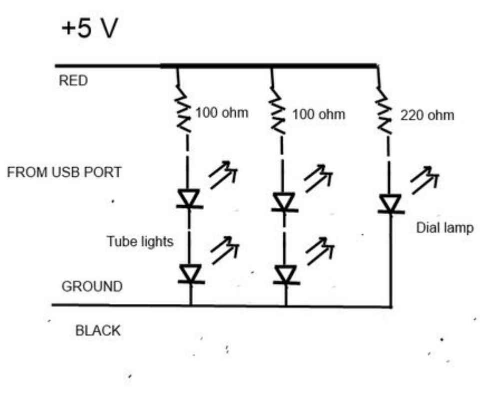
The Bluetooth receiver includes a 1.0 amp USB port. I surfed around for instructables about making a USB-powered light and quickly found out how to have that port power a Superbright yellow LED to shine on the dial from behind. I also created two more circuits in parallel to that one, with two yellow LED's and some small current limiting resistors to create the tube glow effect.

Since I didn't want to damage the tubes, I chose to embed the LED's for the tubes in between two layers of frosted scotch cellophane tape, and then attach those LED/tape assemblies to the back of several tubes with black Gorilla Tape. You don't see that black tape, and the glow effect, although not perfect, makes me want to eventually put this thing where you can see those tubes.





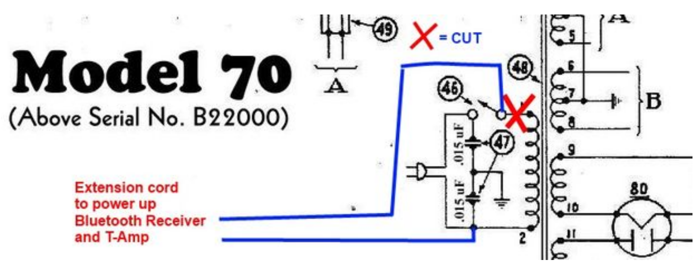




Step 4: Power Switch

Again, out of respect to the radio and the thought that someday someone may wish to restore it fully, I tried to minimize the wiring related to using its on/off switch. Basically, I removed one leg of the main power transformer from use by cutting that wire and taping off the transformer side of it. Then I wired an extension cord into the circuit as shown. This allows the AC on/off switch to interrupt power to the extension cord and totally removes the AC circuit from the radio, with the exception of the small filter capacitors shown in the schematic.

STANDARD WARNINGS of the dangers of working with AC voltage wiring apply here. It's possible to create a short circuit if you don't know what you're doing.



Step 5: Welcome to the 21st century, Philco

The radio is now receiving daily use in our living room. With my cell phone as the source, I can listen to my mp3 music or, even more interesting at times, listen to the world via internet radio using the app TuneIn. I listened to an English language broadcast from Radio Cuba over the weekend, traffic reports in Dublin Ireland and a great oldies program from Chicago.

Improvements that I may make in the future: better lighting for the tubes (maybe orange LED's?), maybe enclosing the speaker. It sounds very, very nice as it is and, to me, beats the heck out of having some Bluetooth brick sitting on the end table.

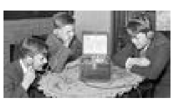
By the way, Thomas Edison had Nipper the dog, but I have Charlie the curious cat, FTW!

Enjoy the video! NOTE: If the video fails to play when clicking on its "PLAY" arrow, try clicking on the title of the video again. That works for me....Brian

Brian

Related Instructables

Bluetooth Tube Radio Project - introduction and opus 1 by yaaaam

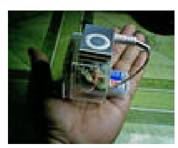


AM DX'ing, the hobby of listening to radio signals from far away... by ke4mcl

Bluetooth tube radio project - opus2 - Japanes pre-war-era's radio Tombstone shape by yaaaam

Re-purposing an Antique Portable Radio Into a Hip Bluetooth Speaker by ke4mcl

how to revive a first generation Olympic 447 portable transistor radio by ke4mcl



how to make mp3 speaker box from cheap radio fm (Photos) by ljuwana

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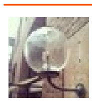
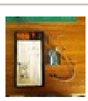


craftclarity says: Jun 18, 2014. 3:07 PM REPLY

I've never heard of those speakers either, they look like they might be handy for a bunch of different applications. Nice!



AlistairM says: Jun 19, 2014. 9:22 AM REPLY

They are rather good, I have a set of 6inch ones. Its great for small audio applications like modifying car stereos for home use, as they have a built in crossover and the smaller speakers handle the higher notes.